

Homework 2

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Exercise 1. Let (X, d) be a metric space. Let $r > 0$, $x \in X$ and define the set

$$B = \{y \in X : d(x, y) \leq r\}.$$

Is it always true that $\text{int}(B) = B_r(x)$? If so, prove it. If not, give a counterexample.

Proof. No. Consider $\mathbb{R}$ under the discrete metric d where $x = 0$ and $r = 1$. Then $B = \mathbb{R}$ and $B_r(x) = \{0\}$. Since every subset of $\mathbb{R}$ is open under d , B itself is open. Thus $B_r(x)$ is not the largest open set contained in B and $B_r(x) \subsetneq \text{int}(B) = B$. $\square$

Exercise 2. Let $C[0, 2\pi]$ be the set of all continuous functions on the interval $[0, 2\pi]$. Let d be a metric on $C[0, 2\pi]$ defined as

$$d(f, g) = \sup_{t \in [0, 2\pi]} |f(t) - g(t)|.$$

Let $f(t) = \sin t$ and $g(t) = \cos t$. Clearly $f, g \in C[0, 2\pi]$. Find the infimum of $r > 0$ such that $f \in B_r(g)$.

Proof. We begin by finding $d(f, g)$. We can do calculus on $f(t) - g(t)$ to get

$$f'(t) - g'(t) = \cos t + \sin t$$

and then solve to get $f'(t) - g'(t) = 0$ when $t = \pi/4$ or $t = 3\pi/4$. In either case, $|f(t) - g(t)| = \sqrt{2}$, so we get that $d(f, g) = \sqrt{2}$. Thus, if $r > \sqrt{2}$, $f \in B_r(g)$, whereas if $r \leq \sqrt{2}$, $f \notin B_r(g)$, so $r = \sqrt{2}$ is the desired result. $\square$

Exercise 3. On $\mathbb{R}^2$, let a, b be two positive real numbers, and define a metric

$$d^{a,b}(\mathbf{x}_1, \mathbf{x}_2) = \sqrt{a^2(x_1 - x_2)^2 + b^2(y_1 - y_2)^2},$$

where $\mathbf{x}_1 = (x_1, y_1)$ and $\mathbf{x}_2 = (x_2, y_2)$. Show that for all $a, b, a', b' > 0$, $d^{a,b}$ and $d^{a',b'}$ are equivalent.

Proof. Let

$$c = \max \left\{ \frac{a}{a'}, \frac{a'}{a}, \frac{b}{b'}, \frac{b'}{b} \right\}.$$

We will prove the squared version of the inequalities, but first note that since $c \geq \frac{a'}{a}$ and $c \geq \frac{b'}{b}$, $a \geq \frac{a'}{c}$ and $b \geq \frac{b'}{c}$. Thus

$$\begin{aligned} \frac{1}{c^2} d^{a',b'}(\mathbf{x}_1, \mathbf{x}_2)^2 &= \frac{1}{c^2} (a')^2 (x_1 - x_2)^2 + \frac{1}{c^2} (b')^2 (y_1 - y_2)^2 \\ &= \frac{(a')^2}{c^2} (x_1 - x_2)^2 + \frac{(b')^2}{c^2} (y_1 - y_2)^2 \\ &\leq a^2 (x_1 - x_2)^2 + b^2 (y_1 - y_2)^2 \\ &= d^{a,b}(\mathbf{x}_1, \mathbf{x}_2)^2. \end{aligned}$$

By similar logic, $ca' \geq a$ and $cb' \geq b$, so

$$\begin{aligned} d^{a,b}(\mathbf{x}_1, \mathbf{x}_2)^2 &= a^2 (x_1 - x_2)^2 + b^2 (y_1 - y_2)^2 \\ &\leq c^2 (a')^2 (x_1 - x_2)^2 + c^2 (b')^2 (y_1 - y_2)^2 \\ &= c^2 d^{a',b'}(\mathbf{x}_1, \mathbf{x}_2)^2. \end{aligned}$$

Now taking square roots (since all the distances are positive), we get the desired inequality showing that $d^{a,b}$ and $d^{a',b'}$ are equivalent. $\square$

Exercise 4. Show that on any metric space, any set A is open if and only if it is a union of open balls.

Proof. First, if A is open, then each point $x \in A$ is an interior point, and thus there exists an open ball B_x centered at x which is contained in A . Then $\bigcup_{x \in A} B_x \subset A$ since each $B_x \subset A$, and $A \subset \bigcup_{x \in A} B_x$ since for each $x \in A$, $x \in B_x \subset \bigcup_{x \in A} B_x$.

Now if A is a union of open balls, then given any $x \in A$, $x \in B_r(y)$ for some $r > 0$ and $y \in A$. By a lemma from class, there exists an $r' > 0$ such that $B_{r'}(x) \subset B_r(y) \subset A$, and thus A is open. $\square$

Exercise 5. On any metric space (X, d) and for any set $A \subset X$, define

$$\delta A = \{x \in X : x \text{ is a boundary point of } A\}.$$

Show that $\delta A = X \setminus (\text{int}(A) \cup \text{int}(A^c))$ where A^c denotes the complement of A .

Proof. The proof is an exercise in restatement. In particular, the set in the last line of the exercise statement can be rewritten as

$$\begin{aligned} X \setminus (\text{int}(A) \cup \text{int}(A^c)) &= \{x \in X : \nexists r > 0 \text{ such that either } B_r(x) \subset A \text{ or } B_r(x) \subset A^c\} \\ &= \{x \in X : \forall r > 0, B_r(x) \cap A \neq \emptyset \text{ and } B_r(x) \cap A^c \neq \emptyset\}. \\ &= \{x \in X : x \text{ is a boundary point of } A\}. \end{aligned}$$

Thus we have $\delta A = X \setminus (\text{int}(A) \cup \text{int}(A^c))$. $\square$